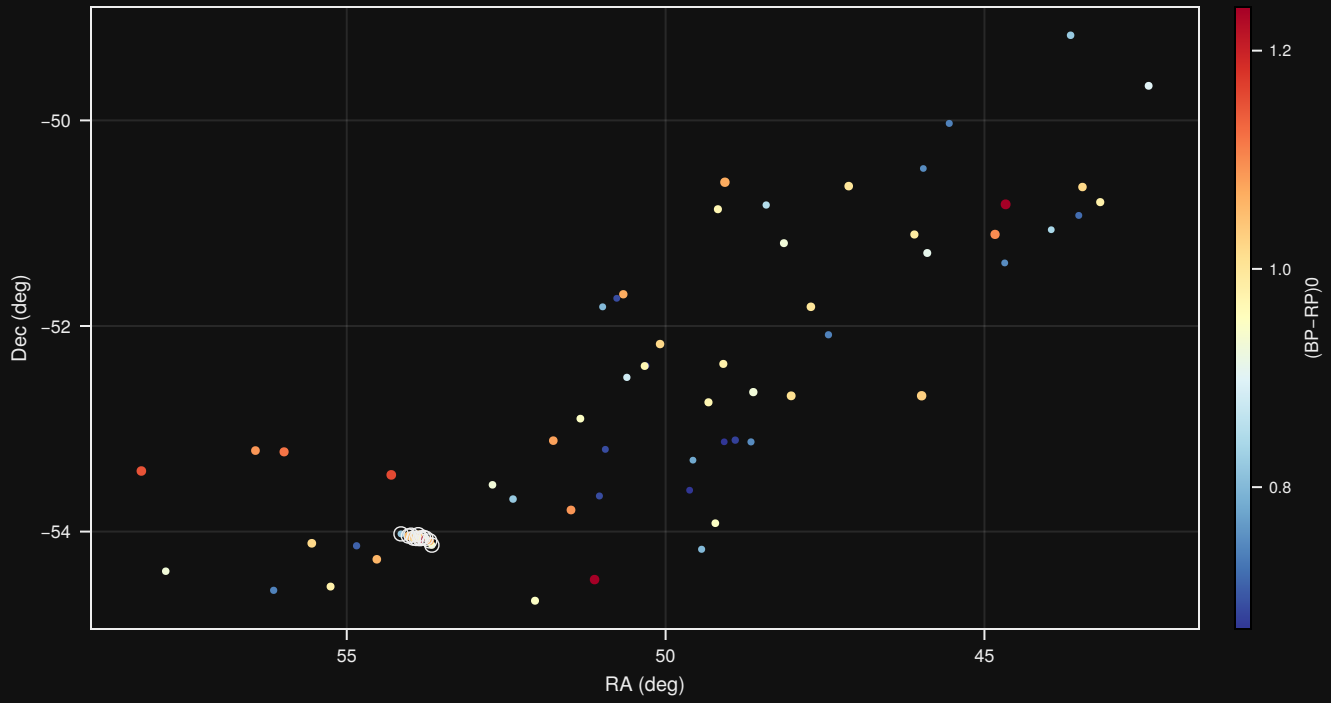
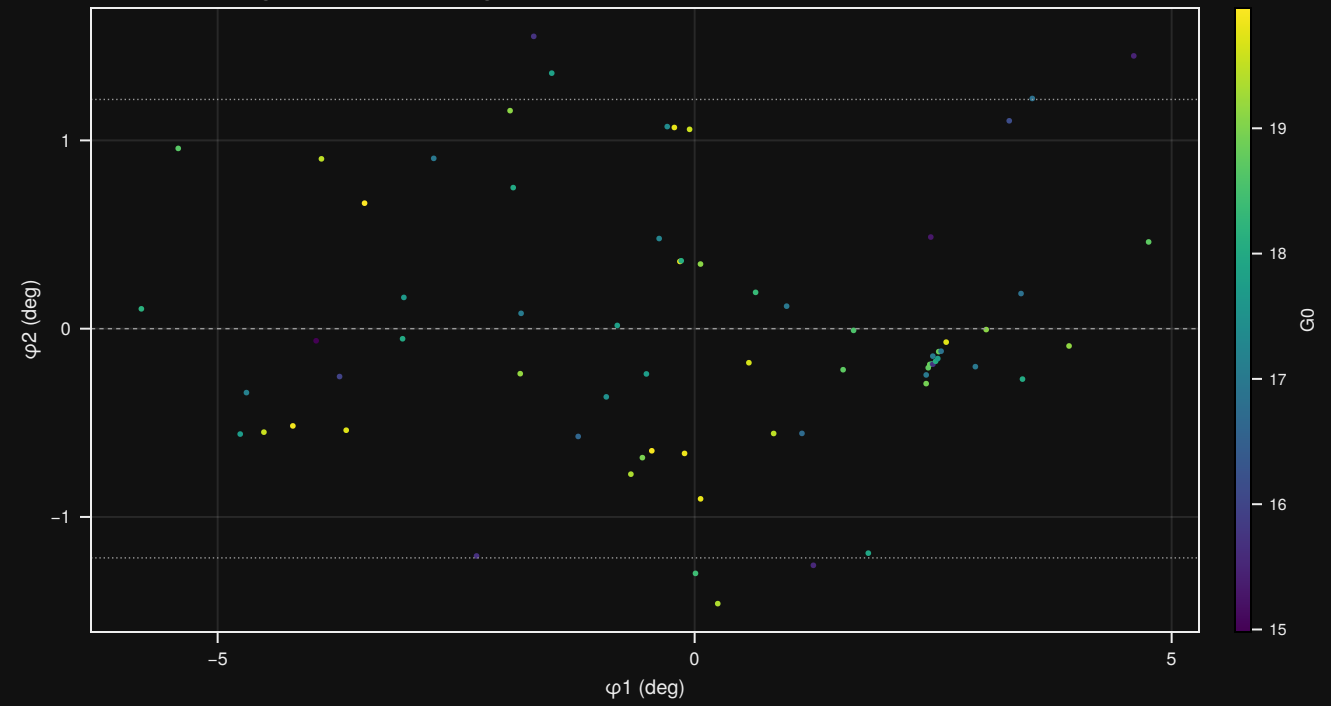
NGC1261b

MEMBERSHIP	
Stars in catalog	70
with Gaia GSP-Phot	44 (63%)
with radial velocity	12
Stream id (sID)	14
GEOMETRY · great-circle fit	
Centroid RA, Dec	50.4, -53.1 deg
Centroid l, b	266.0, -52.3 deg
Pole RA, Dec	84.3, 32.2 deg
Length (φ1, 2–98%)	9.5 deg
Width σ(φ2), robust	0.61 deg
KINEMATICS	
pmRA* med ± σ	2.06 ± 0.45 mas/yr
pmDE med ± σ	-1.54 ± 0.43 mas/yr
μφ1* med ± σ	2.52 ± 0.46 mas/yr
μφ2 med ± σ	-0.28 ± 0.50 mas/yr
Parallax (cat.) med	0.039 ± 0.074 mas
Parallax (wtd, ZP-corr.)	0.052 ± 0.012 mas
Distance	d = 19.3 ± 4.5 kpc
Physical length	3.20 kpc
RV median ± σ(MAD)	63.4 ± 4.4 km/s
RV gradient dV/dφ1	-21.10 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	15.0–20.0 (18.1)
(BP−RP)0 median	0.95
GSP-Phot [M/H] (n=44)	-1.53 ± 1.03
GSP-Phot Teff median	5215 K
RUWE > 1.4	0.0 %
PM clusters (largest)	4 (28 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	70 | 70
n_RV catalog | paper	12 | 12
<[Fe/H]> paper	-1.27
Stellar mass (paper)	5100 Msun
Pole offset vs paper	0.1 deg
Remark	Related to N1261?

Sky map · catalog members (n = 70) · size ∝ brightness · ring = has RV

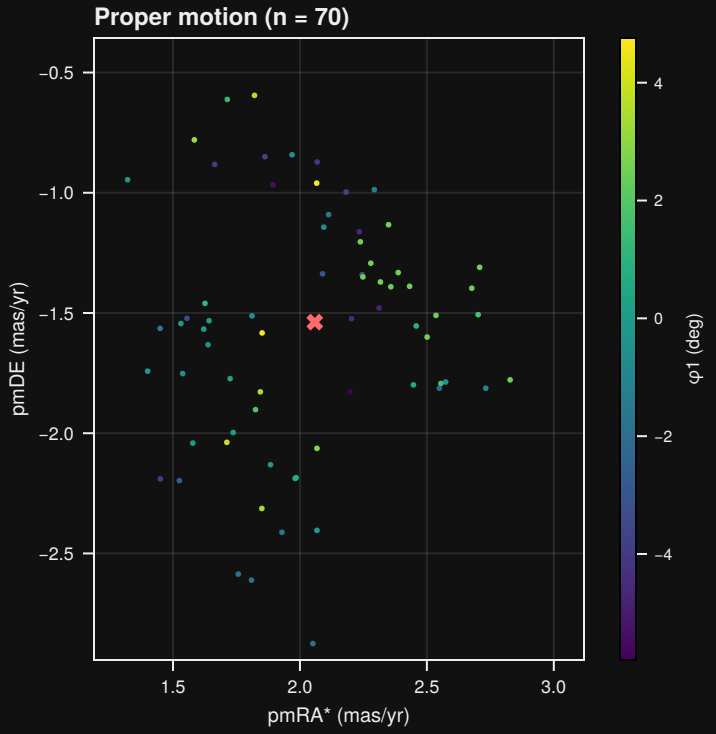


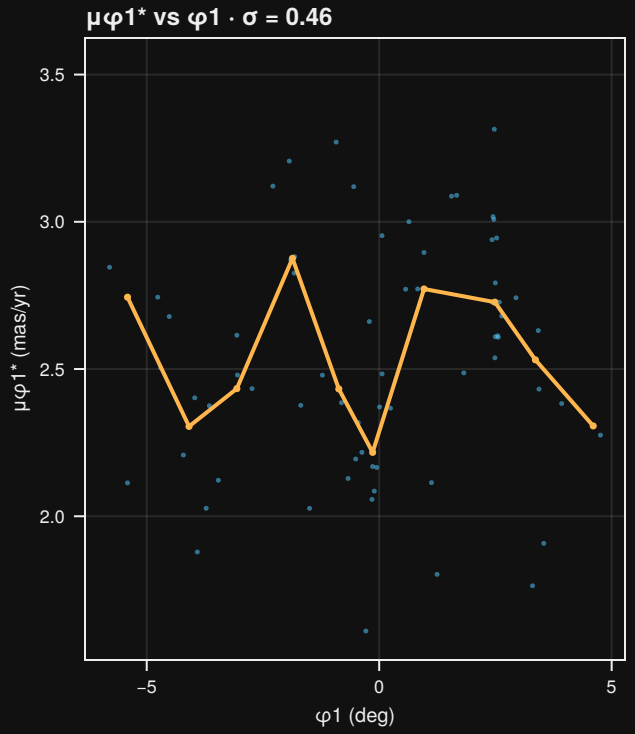
Stream frame (great-circle fit) · length 9.5° · width σ 0.61°

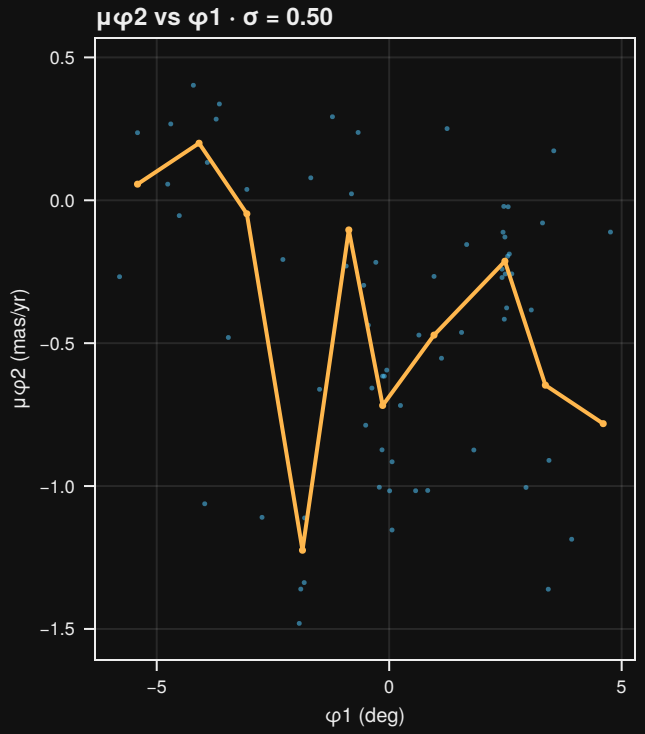


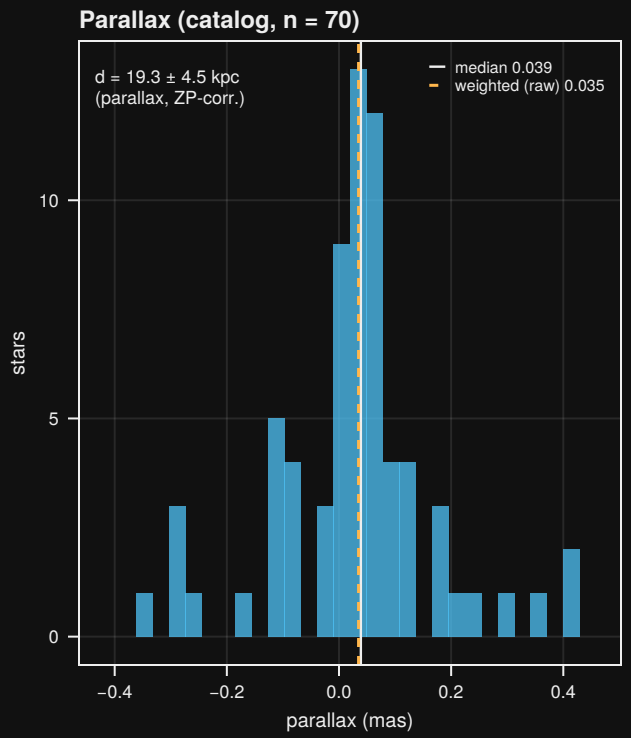
NGC1261b

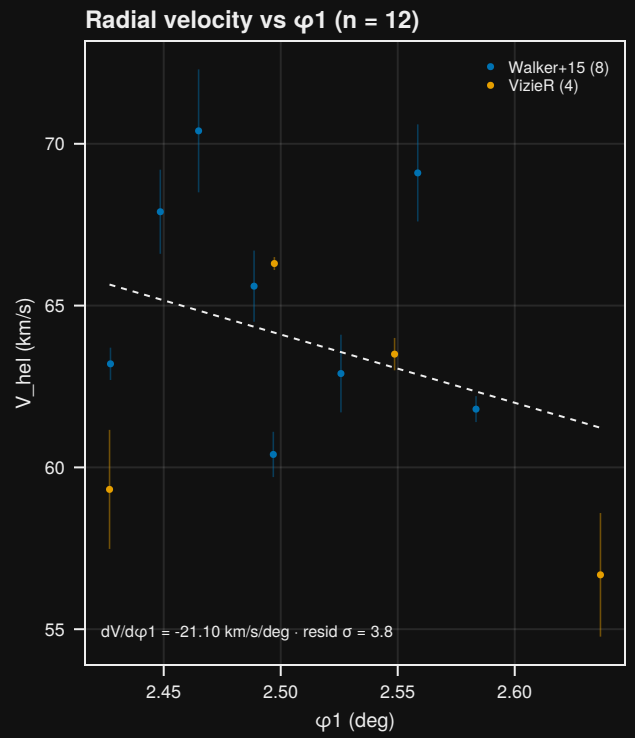
Kinematics · proper motion · parallax · radial velocity

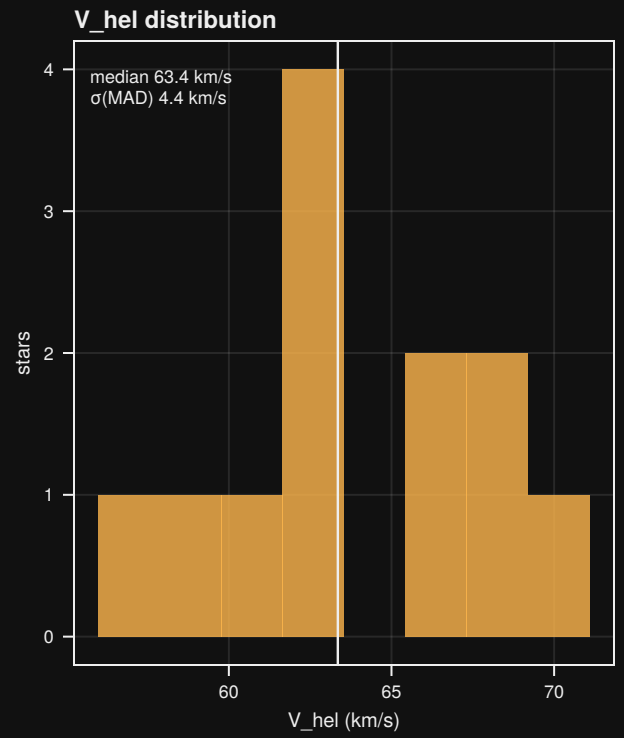






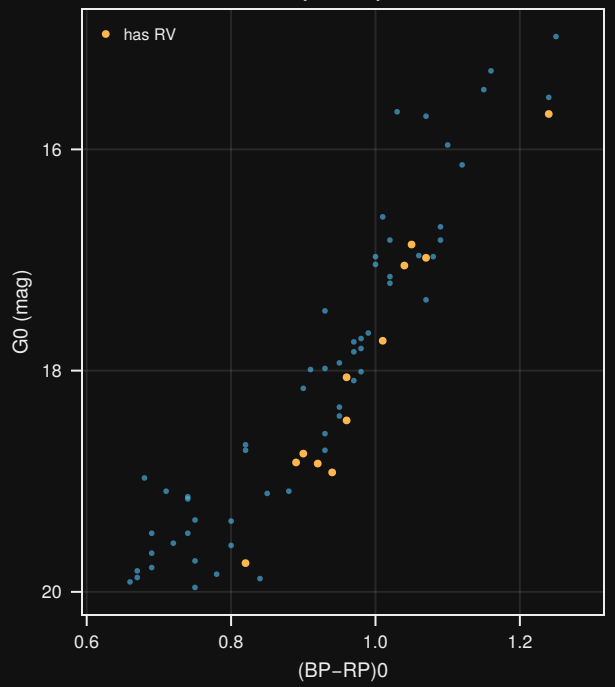




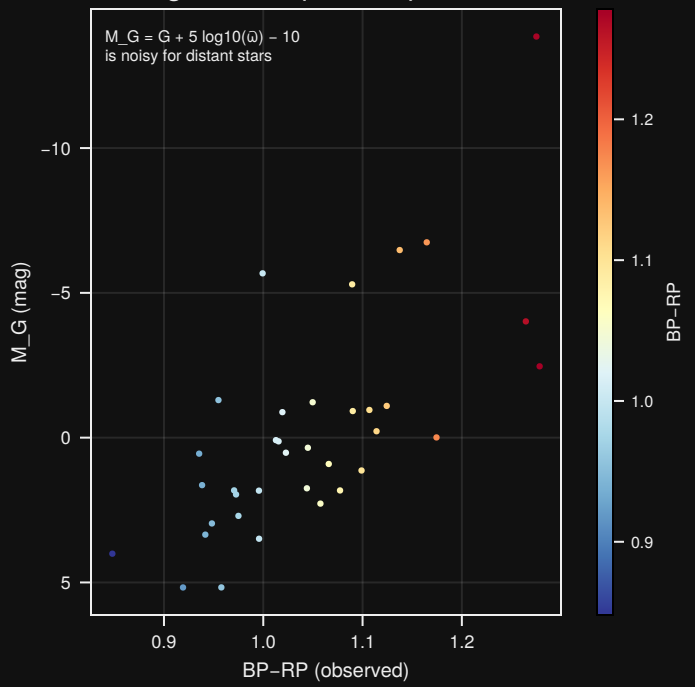


NGC1261b

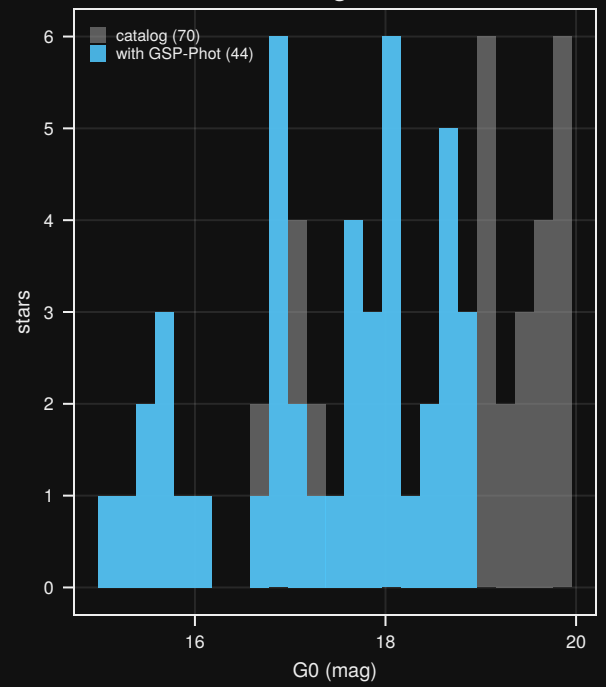
CMD · dereddened (n = 70)



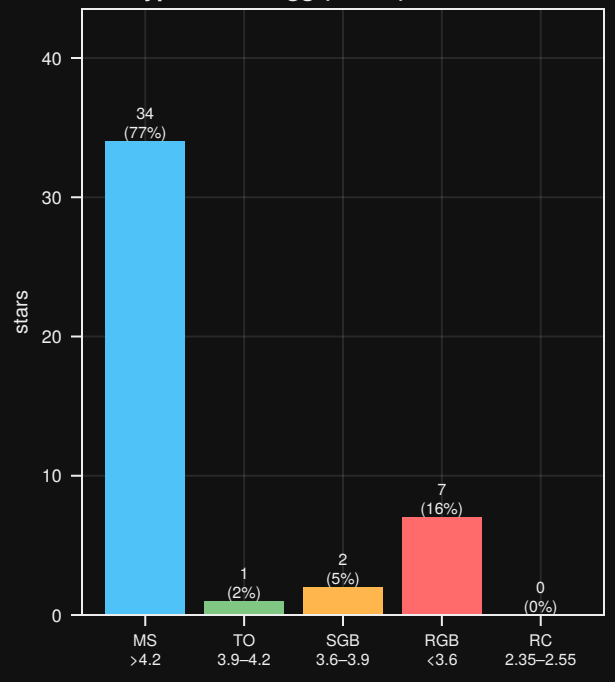
HR diagram · ϖ>0 (n = 36/44)



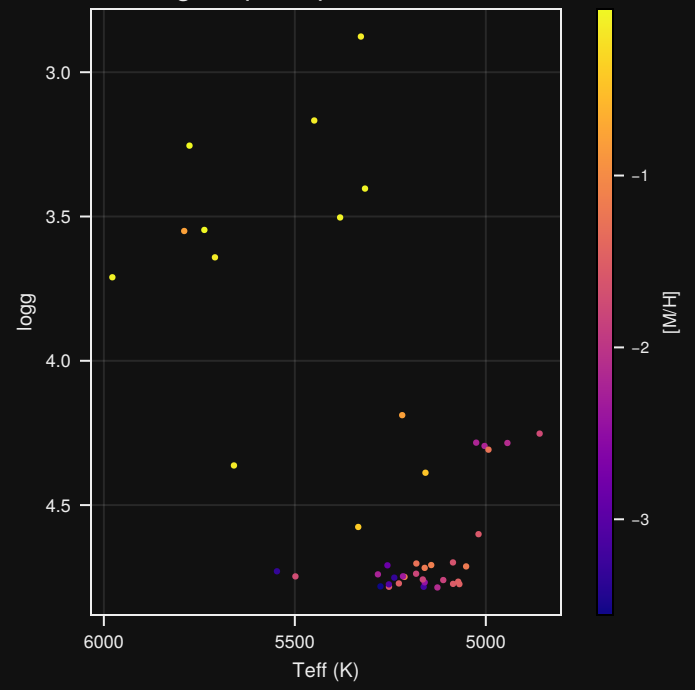
G0 & GSP-Phot coverage



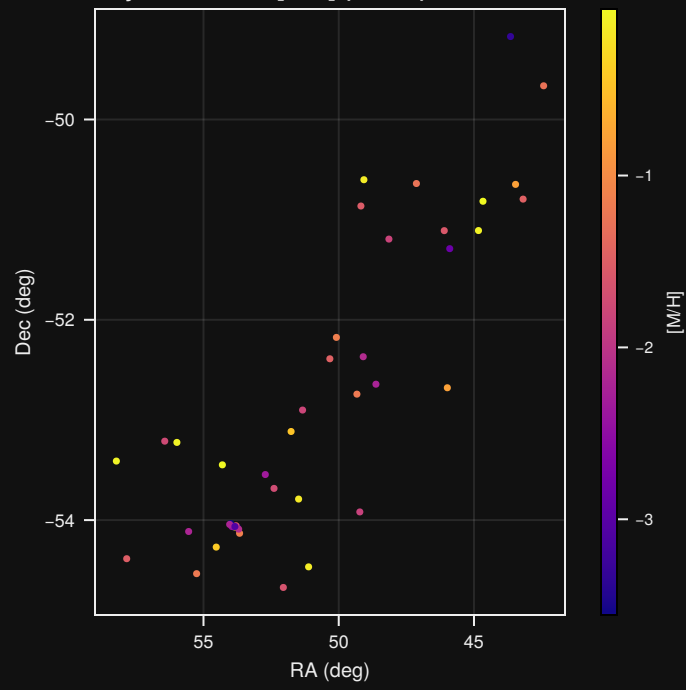
Star types from logg (n = 44)



Kiel diagram (n = 44)



Sky · GSP-Phot [M/H] (n = 44)

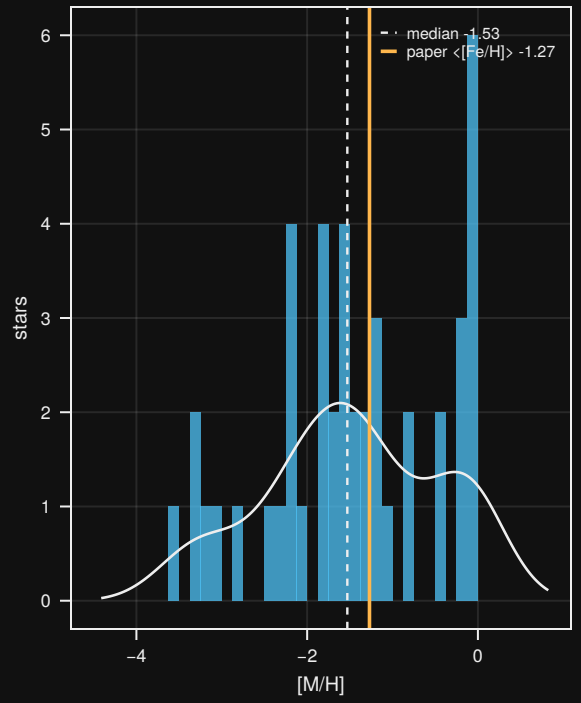


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

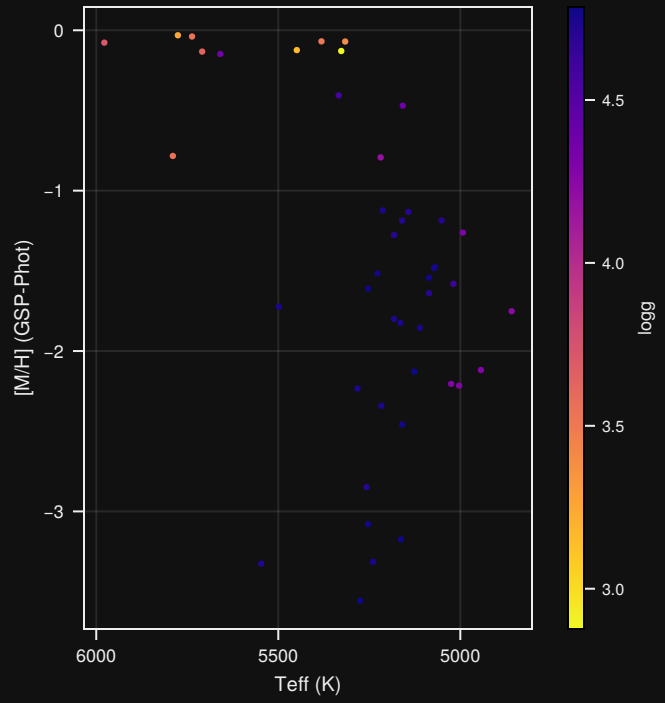
NGC1261b

Metallicity · age · astrometric quality

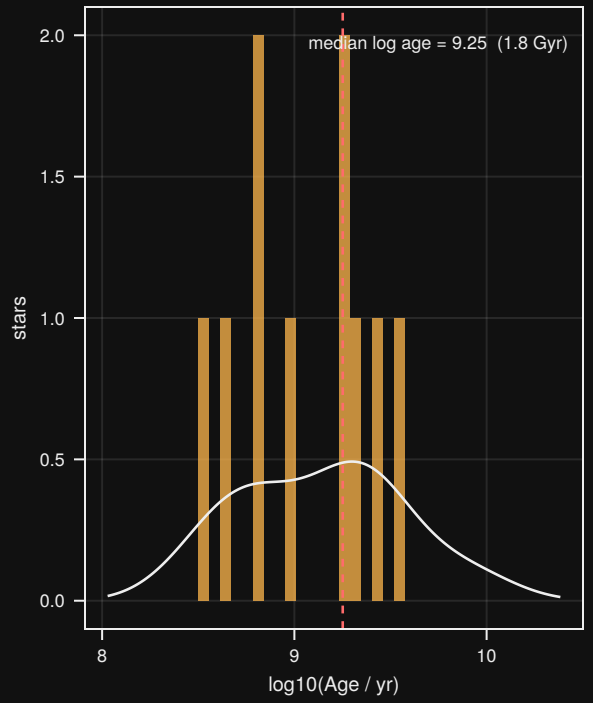
[M/H] GSP-Phot (n = 44)



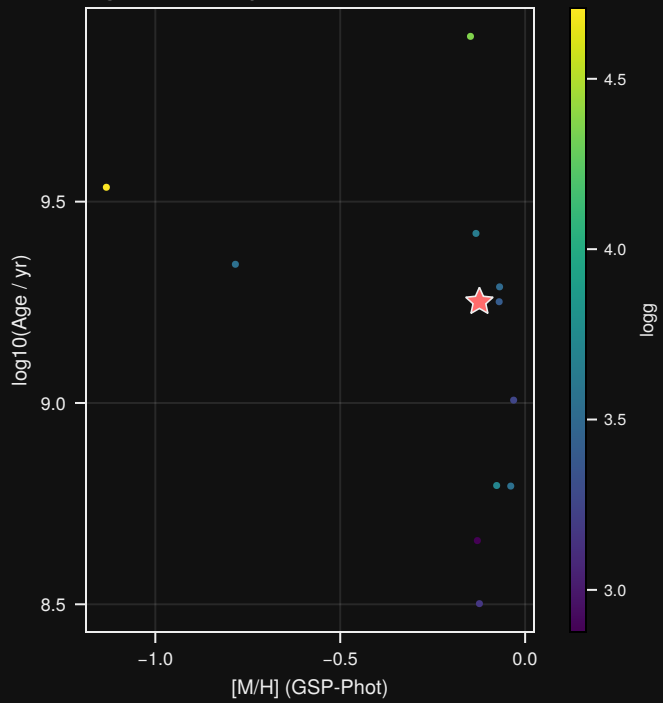
Teff vs [M/H] (n = 44)



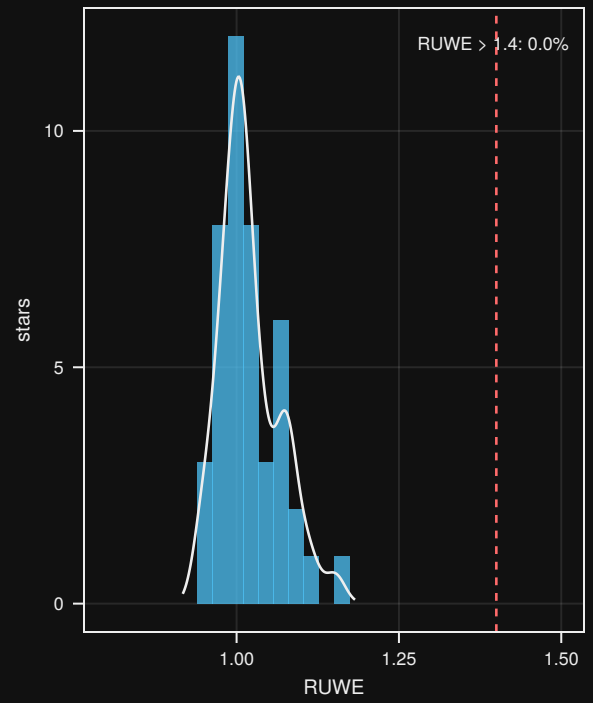
FLAME age (n = 11)



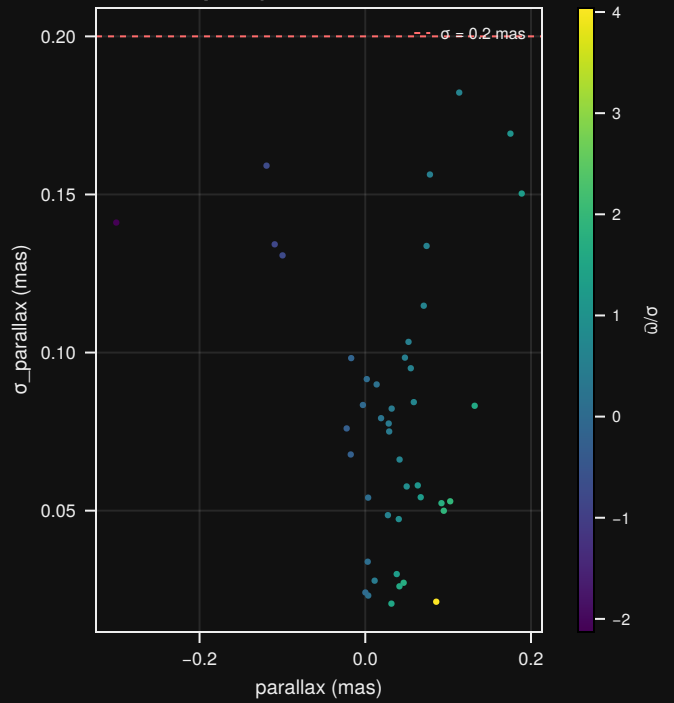
Age–metallicity (n = 11)



RUWE (n = 44)

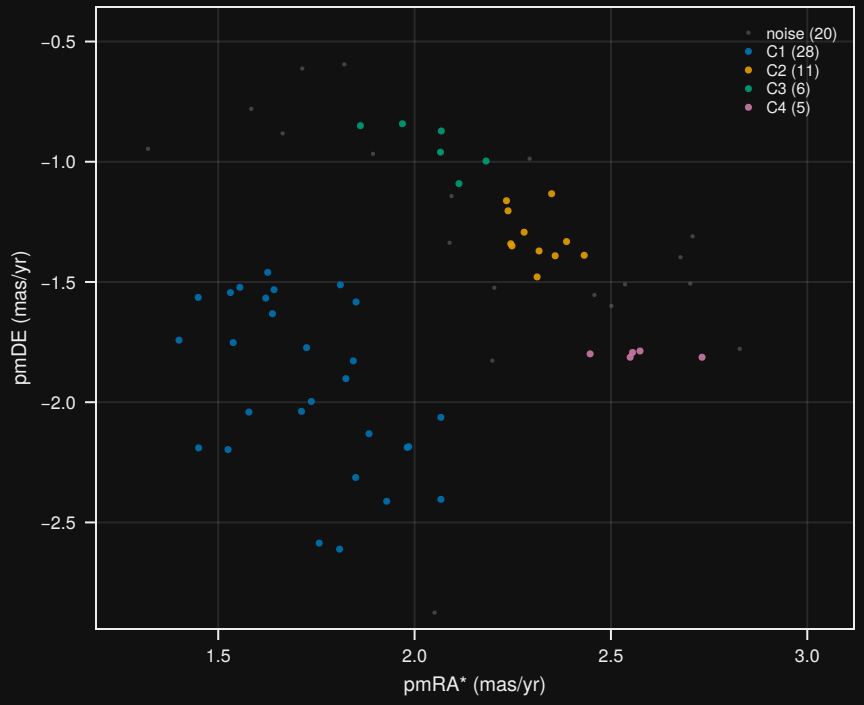


Parallax quality (n = 44)

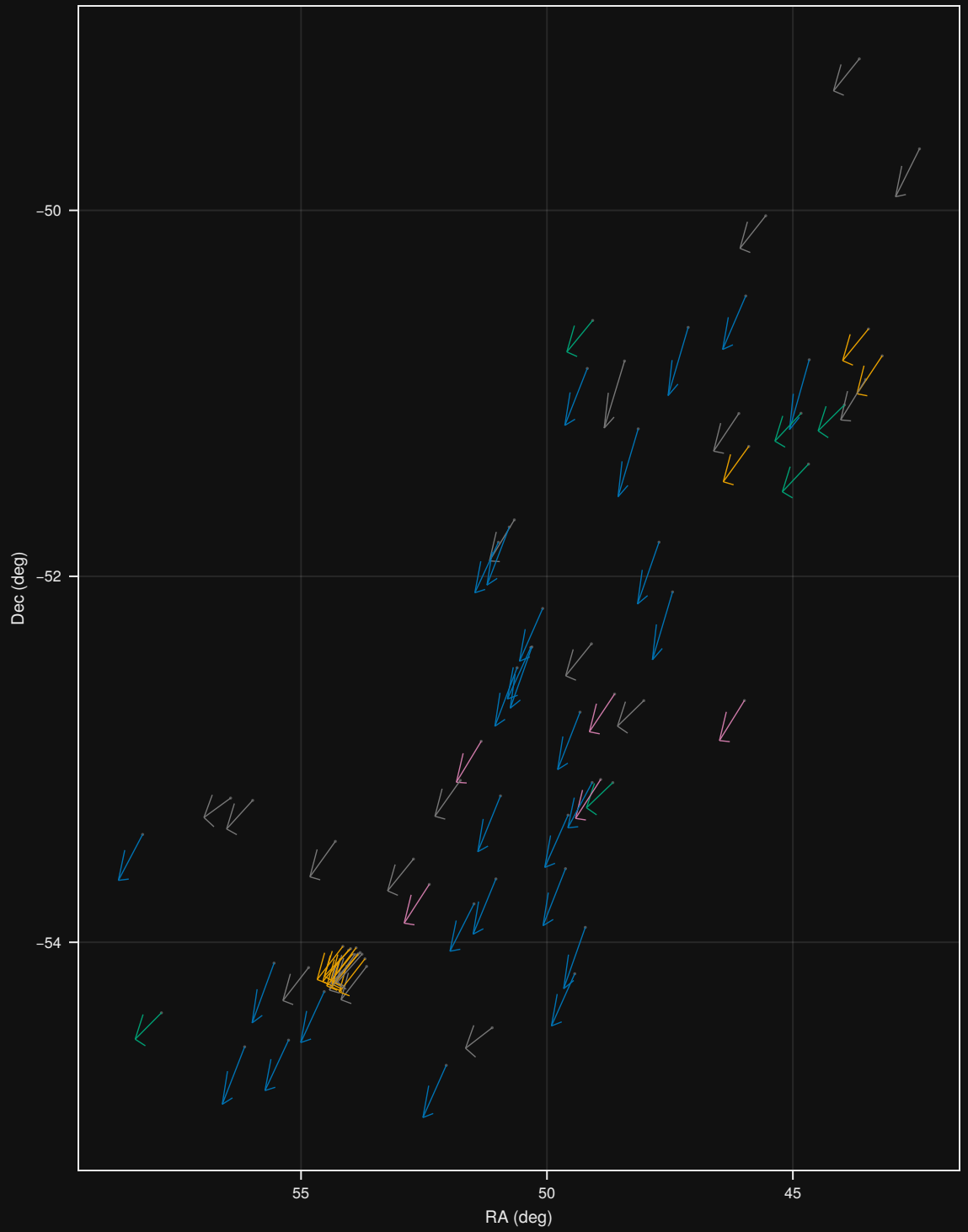


NGC1261b

PM clustering · HDBSCAN · 4 cluster(s) · noise 29%



Proper-motion directions (arrows) coloured by cluster



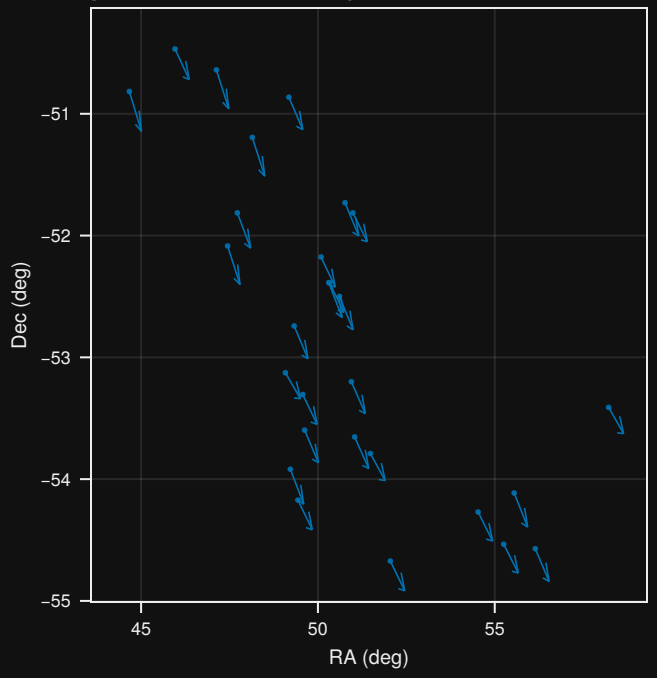
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	20	2.20	-1.37	0.53	-1.60	4
C1	28	1.73	-1.95	0.20	-1.22	0
C2	11	2.31	-1.34	0.09	-2.22	8
C3	6	2.07	-0.92	0.11	-0.13	0
C4	5	2.56	-1.80	0.03	-1.76	0

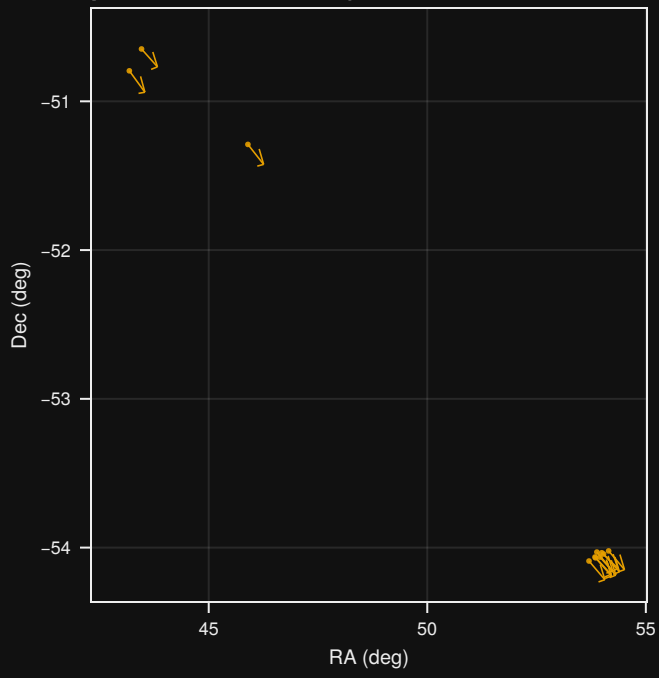
NGC1261b

Per-cluster proper-motion vector maps

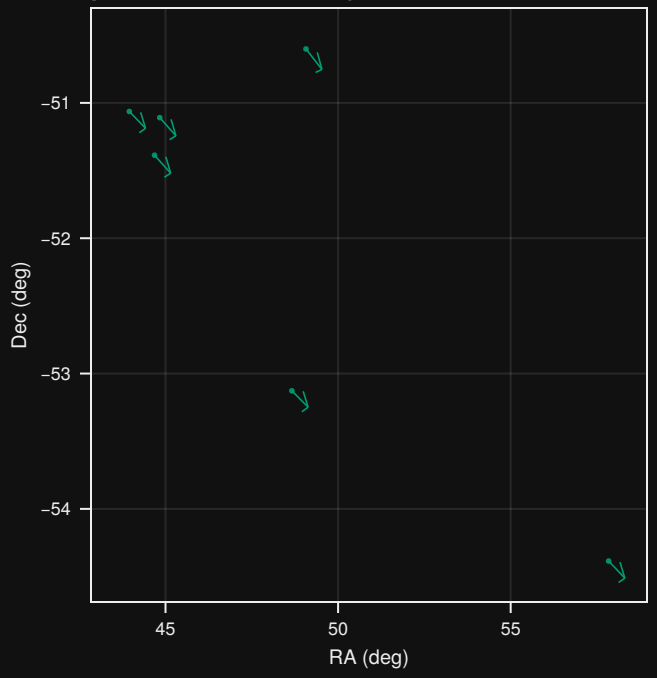
Cluster C1 · n = 28
pm = (1.73, -1.95) mas/yr



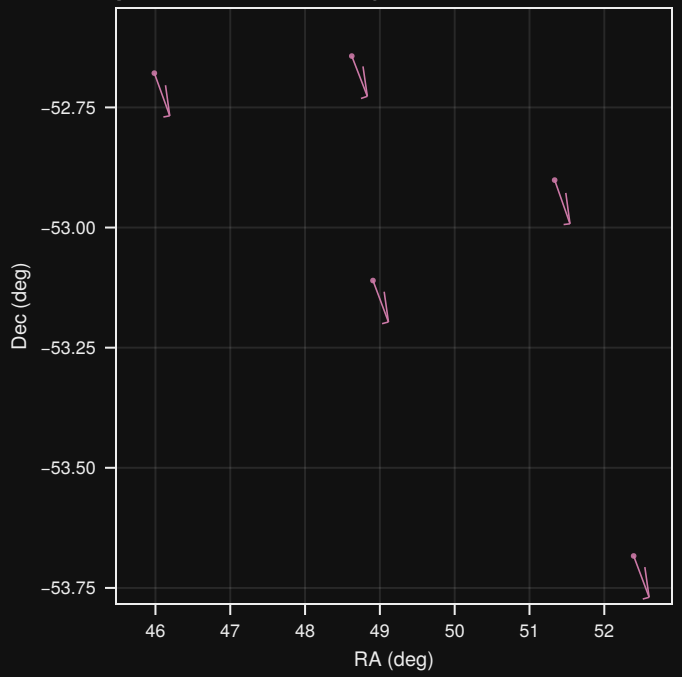
Cluster C2 · n = 11
pm = (2.31, -1.34) mas/yr



Cluster C3 · n = 6
pm = (2.07, -0.92) mas/yr



Cluster C4 · n = 5
pm = (2.56, -1.80) mas/yr



Noise · n = 20

